

Joint Image Enhancement for Underwater Object Detection in Various Domains

Junjie Wen , *Member, IEEE*, Guidong Yang, Benyun Zhao , Lei Lei , Zhi Gao , Xi Chen , *Member, IEEE*, and Ben M. Chen , *Fellow, IEEE*

Abstract—Underwater environments present significant challenges, such as image degradation and domain discrepancies, that severely impact object detection performance. Traditional approaches often use image enhancement as a preprocessing step, but this adds computational overhead, latency, and can even degrade detection accuracy. To address these issues, we propose a novel underwater object detection framework that jointly trains image enhancement within a multitask architecture. This framework employs a progressive training strategy to iteratively improve detection performance through enhancement and introduces a domain-adaptation mechanism to align features across domains at both image and object levels. Experimental results demonstrate that our method achieves state-of-the-art performance across diverse data sets, with real-time detection at 105.93 frames per second and over +15% mean average precision absolute improvement in unseen environments, underscoring its potential for real-world underwater applications.

Index Terms—Autonomous underwater vehicle (AUV), domain adaptation, underwater image enhancement (UIE), underwater object detection (UOD).

I. INTRODUCTION

RECENT advances in deep learning-based object detection [1], [2], [3] have significantly propelled progress across various fields. However, underwater object detection (UOD) faces unique challenges that have slowed its development. The first challenge lies in the complex underwater environment, where poor illumination, color distortion, and blurring result in low-quality images and domain discrepancies, complicating detection tasks. The second challenge stems from the limited computational resources of autonomous underwater

vehicles (AUVs), which further constrain the implementation and performance of UOD systems.

Several UOD methods adopt a direct pipeline [see Fig. 1(a)] to detect objects from degraded underwater images [4], [5]. However, these methods do not address the degraded image quality, which can impair UOD performance. To tackle this issue, some researchers incorporate underwater image enhancement (UIE) techniques to improve detection accuracy. Among these, a sequential pipeline [see Fig. 1(b)] is often employed, where UIE serves as a preprocessing step prior to UOD [6], [7], [8]. However, such approach significantly increases inference latency, posing challenges for real-time underwater detection. Furthermore, recent studies have shown that using UIE as a preprocessing step may fail to improve, or even degrade UOD performance [9], [10], [11]. While sequential architectures are commonly used, other approaches propose either parallel architectures [see Fig. 1(c)] or multitask architectures [see Fig. 1(d)] to jointly train UOD and UIE tasks in an end-to-end manner [12], [13], [14]. However, these designs often introduce additional network structures to connect UIE and UOD tasks, which not only increase model complexity but also exacerbate inference latency, limiting their practicality for resource-constrained AUVs.

Moreover, training an effective detector that integrates an enhancement task in an end-to-end manner poses significant challenges. The enhancement task primarily aims to improve the visual quality of underwater images by enhancing details, while the detection task focuses on identifying and localizing object features. Although some studies introduce additional network structures to achieve this balance [14], [15], these methods increase both computational resources and inference latency, which restricts their practicality in resource-constrained environments. To the best of the authors' knowledge, no existing work has achieved a balance between these two tasks without relying on additional network structures.

Furthermore, the variability and complexity of underwater imaging conditions often lead to significant domain shifts across different environments, making it challenging for networks to adapt to unfamiliar settings. While domain-adaptation techniques have been explored for UIE [16], [17], [18], their application in UOD remains largely underexplored. Although domain adaptation strategies have demonstrated success in object detection underwater adverse terrestrial conditions [19], [20], extending these methods to underwater scenarios is particularly challenging due to the difficulty of obtaining high-quality underwater images.

Received 7 January 2025; revised 30 May 2025; accepted 31 July 2025. Date of publication 30 October 2025; date of current version 23 January 2026. The work was supported in part by the Research Grants Council of Hong Kong SAR under Grant 14206821, Grant 14217922, and Grant 14209623, in part by The InnoHK of the Government of the Hong Kong SAR via Hong Kong Centre for Logistics Robotics, in part by The Talent Program of Guangdong Province under Grant 2021QN02Z107, in part by the National Science and Technology Major under Project 2024ZD01NL00103, and in part by The Major Key Project of PCL under Grant PCL2025A03. (Corresponding author: Lei Lei.)

Associate Editor: C. Guo.

Junjie Wen is with Pengcheng Laboratory, Shenzhen 518000, China, and also with the Department of Mechanical and Automation Engineering, Chinese University of Hong Kong, Hong Kong (e-mail: jjwen@mae.cuhk.edu.hk).

Guidong Yang, Benyun Zhao, Lei Lei, Xi Chen, and Ben M. Chen are with the Department of Mechanical and Automation Engineering, Chinese University of Hong Kong, Hong Kong (e-mail: leilei01@cuhk.edu.hk).

Zhi Gao is with the School Remote Sensing and Information Engineering, Wuhan University, Wuhan 430072, China.

Digital Object Identifier 10.1109/JOE.2025.3604170

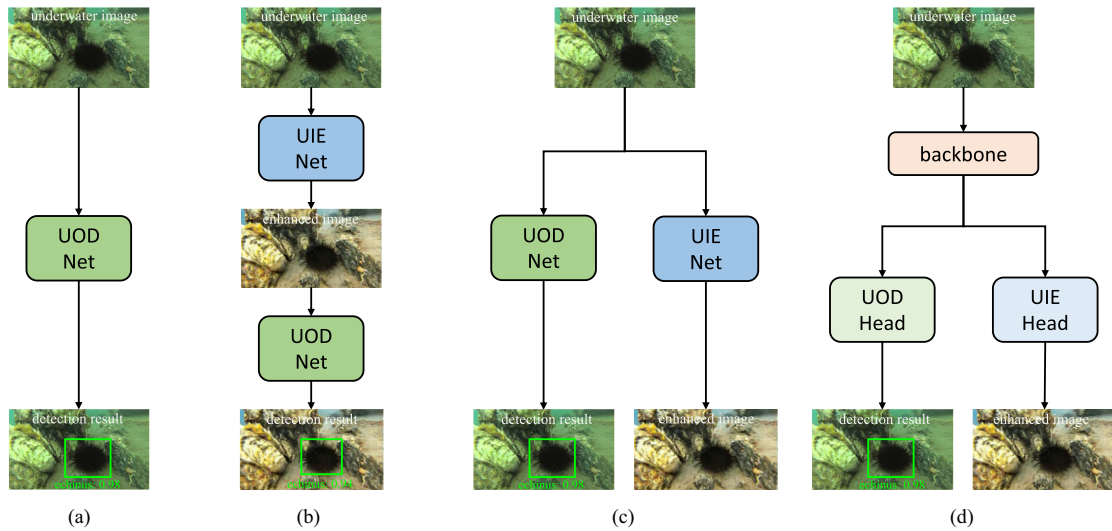


Fig. 1. Different architecture designs for UOD. (a) *Direct* design that directly detects underwater objects. (b) *Sequential* design that uses the UIE as a preprocessing step. (c) *Parallel* design that uses UIE network to assist UOD performance. (d) *Multitask* design where UOD and UIE heads share the same backbone.

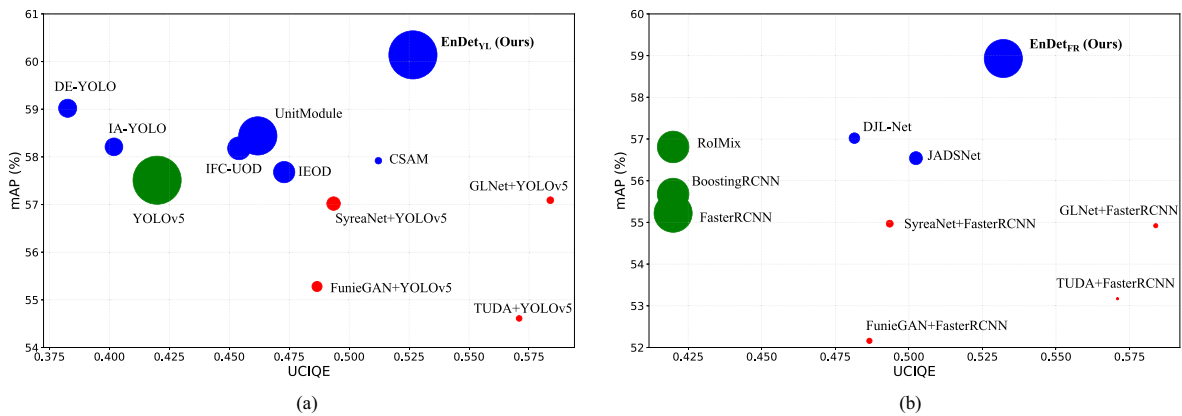


Fig. 2. Performance comparison of various UOD methods in both detection and enhancement tasks using the DUO [21] test set. The y-axis represents the mAP, indicating detection performance, while the x-axis represents UCIE [22], measuring enhancement performance (higher values indicate better enhancement). The size of each dot corresponds to the inference speed fps, with larger dots denoting faster speeds. Blue dots represent Joint-UIE-UOD methods, green dots represent Origin-then-UOD methods, and red dots represent UIE-then-UOD methods. (a) One-stage Detector. (b) Two-stage Detector.

To address the above challenges, we present EnDet, an innovative framework for UOD that jointly trains image enhancement and object detection within a unified multitask architecture. Unlike existing methods that rely on additional network structures to balance UOD and UIE tasks, EnDet adopts a progressive training strategy to iteratively improve the performance of both tasks. In addition, it incorporates a novel domain-adaptation mechanism to ensure robust performance across diverse underwater conditions. Extensive experiments demonstrate that EnDet not only enhances underwater image quality and detection accuracy but also achieves real-time inference, showcasing its potential for practical underwater exploration (as shown in Fig. 2). The key contributions of this work include the following.

1) We propose a unified *multitask* architecture that *jointly* trains both UOD and UIE. This design enables the removal of the UIE task head during inference without affecting UOD performance, thereby significantly reducing computational cost and latency.

- 2) A progressive training strategy is introduced to iteratively improve the performance of both detection and enhancement tasks *without relying on additional network structures* to balance UOD and UIE.
- 3) A novel domain-adaptation mechanism is developed to address domain shifts across diverse underwater environments, enhancing detection robustness and ensuring consistent performance in unfamiliar conditions.
- 4) Extensive experimental results demonstrate that the proposed framework outperforms state-of-the-art (SOTA) UOD methods across various data sets, achieving significant improvements in both new environments and real-time inference performance.

The rest of this article is organized as follows. Section II provides an overview of prior related research. Section III presents our proposed framework in detail. In Section IV, extensive experiments are conducted across various UOD data sets to evaluate the performance of our method against other SOTA

approaches. In addition, Section V discusses the potential limitations of our method by assessing the impact of selecting different synthetic underwater images. Finally, Section VI concludes this article and offers insights into future research directions in the field of underwater vision tasks.

II. RELATED WORK

A. General Object Detection

General object detection aims to identify objects of interest within an image and accurately determine their respective categories and positions [1], [3]. The approaches employed for this purpose can be primarily classified into two types: two-stage detectors and one-stage detectors.

Two-stage detectors operate through an initial stage that generates region proposals potentially containing objects, which are then refined to accurately classify and locate the objects within bounding boxes. Introduced in 2014 by Girshick et al. [23], R-convolutional neural network (CNN) was a pioneering object detection method that combined high-capacity CNNs with selective search for region proposal. In 2015, Girshick [24] introduced Fast R-CNN, which improved the efficiency of R-CNN by sharing CNN computations across all region proposals. Ren et al. [1] then enhanced the detection process by introducing faster R-CNN, which integrates a fully convolutional network that predicts object boundaries and scores with the main CNN to share computed features, significantly improving detection performance.

Compared to two-stage detectors, one-stage detectors optimize processing speed by eliminating the region proposal stage and directly predicting bounding boxes and class probabilities across the image in a single pass. For example, Redmon et al. [2] proposed YOLO in 2016, a revolutionary approach that processes the entire image in a single network pass to directly predict bounding box coordinates and class probabilities. Liu et al. [25] introduced SSD that uses multiple feature maps at different scales to predict bounding boxes and class scores, enhancing its effectiveness in detecting smaller objects compared to YOLO. In recent years, YOLO has continued to evolve with various versions [3], [26] that focused on both speed and accuracy improvements while integrating advanced techniques.

In this work, we adapt our framework to work with both the one-stage detector YOLOv5 [3] and the two-stage detector faster R-CNN [1], demonstrating its versatility across different types of detectors.

B. Underwater Object Detection

Significant progress has been made in object detection techniques in terrestrial environments [1], [2], [3]. However, the complexity of underwater environments, which often degrades image quality, presents considerable obstacles for UOD in such scenarios.

The architecture designs for UOD can be categorized into the following four types.

1) *Direct*: In the direct design [see Fig. 1(a)], degraded underwater images are directly input into detectors, with modifications

to general detectors to enhance UOD performance. RoIMix [4] used augmentation to simulate overlapping and blurred objects, enhancing UOD performance in underwater environments. Boosting R-CNN [5] introduced a two-stage method with probabilistic inference and boosting reweighting for refinement. While straightforward, this design's reliance on degraded training images highlights significant room for improvement.

2) *Sequential*: Sequential-designed UOD methods [see Fig. 1(b)] first enhance degraded underwater images with a UIE network, followed by processing with a UOD network. For example, Jiang et al. [6] used WaterNet [27] to reduce haze and color cast, introducing a CSAM module to combine original and enhanced feature maps for improved detection. Yeh et al. [7] proposed a lightweight method with an image color conversion network to adjust color information before detection. Liu et al. [8] introduced UnitModule, which employs unsupervised learning losses, a color cast predictor, and underwater color transfer augmentation to enhance detection performance. However, sequential designs often introduce additional computational overhead and latency, as the UOD network must wait for the UIE process during inference.

3) *Parallel*: Parallel frameworks [see Fig. 1(c)] use separate networks for UOD and UIE tasks, addressing detection and enhancement simultaneously. For example, Wang et al. [13] proposed a dual-branch network with image decolorization and edge enhancement to improve feature fusion, but its tightly coupled branches increased network complexity. Dai et al. [28] proposed a UOD network with cross-domain interaction for complementary information and gated fusion for adaptive control. Liu et al. [12] used UIEC2Net [29] for UIE and introduced a removable feature-guided module to refine UOD feature maps for efficiency. However, reliance on existing UIE methods optimized for visual enhancement may limit improvements in UOD performance.

4) *Multitask*: A multitask design [see Fig. 1(d)] trains an underwater object detector using a shared backbone for UOD and UIE tasks, with separate task-specific heads. For instance, Cheng et al. [14] proposed JADSNet, which combines image enhancement and object detection in a unified framework with end-to-end training. However, its enhancement references rely on existing UIE methods [30] optimized for visual quality, which may not effectively support detection. Furthermore, interconnection of task heads through additional structures increases computational costs and testing latency.

Based on different training strategies, UOD methods can be categorized into the following three types.

1) *Origin-Then-UOD*: This approach applies object detection directly to degraded underwater images without enhancement or preprocessing. Methods in this category focus on adapting general detection architectures to handle degraded inputs [4], [5]. While simple and efficient, their performance is often limited by the poor quality of underwater images.

2) *UIE-Then-UOD*: In this approach, a UIE network is first trained to enhance degraded underwater images, and the enhanced outputs are then used to train the object detector. While this approach aims to improve detection by mitigating issues like haze and color distortion, it often increases computational

latency. Moreover, since the UIE network is typically optimized for visual quality rather than detection-specific features, it can sometimes degrade detection performance instead of enhancing it [9], [10], [11].

3) *Joint-UIE-UOD*: This approach jointly trains UIE and UOD tasks in an end-to-end manner. While this approach aims to optimize both tasks simultaneously, they face notable limitations. Some [6], [12], [14] rely on existing UIE methods, resulting in sub-optimal UOD performance. Others [7], [8], [13], [28] introduce additional network structures to connect UIE and UOD tasks, increasing model complexity and inference time, thereby limiting their suitability for real-time applications.

In this work, we propose a *multitask* UOD framework that enables the *joint* training of UIE and UOD tasks. Unlike existing methods that rely on additional network structures, we introduce a novel joint training strategy to effectively balance the performance of UIE and UOD, significantly reducing both model complexity and inference latency.

C. Underwater Image Enhancement

UIE techniques are considered crucial for high-level underwater vision tasks as the captured images are degraded by complex underwater environments [31], [32]. Unlike traditional methods [33] that rely on prior information to recover clear images, recently developed learning-based methods directly transform degraded underwater images into clear ones. For example, Fu and Cao [32] proposed a two-branch network to address global color distortion and local contrast reduction, resulting in a lightweight architecture that achieves SOTA performance. Zhang et al. [34] introduced a UIE framework that leverages light field and depth map optimization, and dual encoding sub-networks to achieve superior color correction and visual quality in underwater images. Huang et al. [35] proposed SemiUIR, a mean-teacher framework that incorporates unlabeled data into network training and introduces a reliable bank for storing pseudoreference images to effectively improve underwater image quality. Wen et al. [36] proposed a computationally efficient transformer-based network featuring a learnable environment adaptor, enabling the network to adapt effectively to various environments. Xiao et al. [37] proposed a stochastic resonance-based UIE method for enhancing turbid underwater images, which improves visual quality by leveraging adaptive parameter optimization and evaluations on multiple data sets. However, recent studies [9], [10], [11] have revealed that UIE can adversely affect high-level underwater vision tasks like object detection, as current UIE methods are primarily designed for visual enhancement rather than task-specific performance. Therefore, in this study, we explore the potential of effectively combining UIE with high-level UOD tasks.

D. Underwater Domain Adaptation

Domain adaptation aims to reduce distribution differences across multiple domains and has been applied to a variety of tasks [38]. In the field of underwater vision, domain adaptation is extensively employed in UIE to handle the diverse and complex degradation effects resulting from varying underwater conditions. For example, Chen and Pei [16] proposed a domain

adaptation framework for UIE that separates encoded features into content and style latents to achieve SOTA performance in various real-world benchmarks. Wang et al. [18] introduced TUDA that minimizes interdomain and intradomain gaps through a triple-alignment network for adversarial learning and a rank-based underwater quality assessment method for easy-hard adaptation. Wen et al. [17] proposed a framework for UIE that integrates synthetic and real data using a revised underwater formation model and novel interdomain and intradomain adaptation strategies. However, these advancements are primarily focused on UIE, leaving domain adaptation for UOD largely unexplored. Although Liu et al. [39] introduced a domain generalization technique using WCT2 style transfer [40], their approach focused on enhancing underwater object detectors by diversifying input images, rather than tackling domain shift issues. While domain adaptation has proven successful in terrestrial object detection [19], [20], its effectiveness in underwater scenarios is limited by the challenge of obtaining clear images. In this study, we introduce a novel domain-adaptation strategy for UOD by effectively leveraging the enhanced image domain at different feature levels.

III. PROPOSED METHOD

A. Framework Overview

The objective of our proposed method is to enhance UOD performance in various underwater environments by integrating the UIE task head in a joint training approach, enabling our network to effectively facilitate both UOD and UIE simultaneously. As illustrated in Fig. 3, the framework employs a multitask architecture where the backbone is shared by both the UIE and UOD heads. Unlike existing methods that rely on additional network structures to balance UOD and UIE, our approach achieves this balance through a progressive training strategy, avoiding increased model complexity and inference latency.

Moreover, to facilitate a domain-adaptive training, we introduce three distinct image data domains: the original domain $\mathcal{D}_o$, the synthetic domain $\mathcal{D}_s$, and the enhanced domain $\mathcal{D}_e$. The original domain consists of images from real-world underwater images, the synthetic domain includes images generated through synthetic underwater imaging techniques, and the enhanced domain comprises images enhanced from the original domain.

B. Progressive Training Strategy

The structure of the proposed framework is straightforward; however, integrating a UIE task head to enhance the UOD task presents challenges, as UIE focuses on improving visual quality, whereas UOD aims to identify and localize object features [10]. Unlike existing methods that introduce additional network structures to balance UOD and UIE, our framework achieves this balance through a progressive training strategy (see Algorithm 1), which reduces both model complexity and inference latency. The proposed training strategy consists of two stages: the burn-in stage and the mutual-learning stage.

1) *Burn-In Stage*: In this stage, the objective is to equip the network with fundamental capabilities necessary for both UOD and UIE tasks. Given the distinct objectives of UOD and UIE, simultaneous integration of these functionalities might lead to

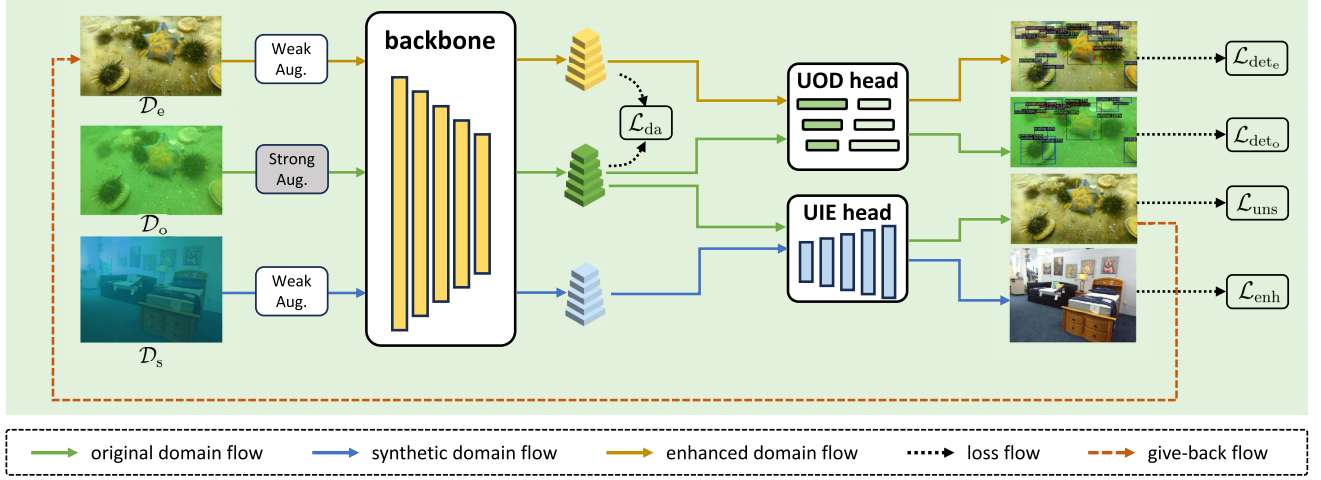


Fig. 3. Overview of our proposed EnDet. Our framework leverages a multitask architecture to jointly train both UOD and UIE tasks, which is achieved through a progressive training strategy. We also introduce three distinct image data domains: the original domain $\mathcal{D}_o$, the synthetic domain $\mathcal{D}_s$, and the enhanced domain $\mathcal{D}_e$.

Algorithm 1: Pseudocode for Progressive Training Strategy.

Input: Images from $\mathcal{D}_o$ and $\mathcal{D}_s$; Burn-In epochs E_b ; Maximum epochs $E_{\max}$; Training step n ; Phase change step n_c ; Phase flag $f \in \{0, 1\}$;
Output: Final network $\Theta^{E_{\max}}$.

```

1: Initialize network  $\Theta^0$ ;  $n \leftarrow 0$ 
2: for  $e \leftarrow 0$  to  $E_{\max}$  do
3:    $f \leftarrow 0$ 
4:   if  $e \leq E_b$ ; do # Burn-In stage
5:     for each epoch do
6:        $n \leftarrow n + 1$ 
7:       if  $f = 0$ ; do # BID phase
8:         Minimize  $\mathcal{L} \leftarrow \mathcal{L}_{\text{det}_o}$ 
9:       else # BIE phase
10:        Minimize  $\mathcal{L} \leftarrow \mathcal{L}_{\text{enh}}$ 
11:       if  $\text{mode}(n, n_c) = 0$ ; change  $f$ 
12:     else
13:       for each epoch do # Mutual-Learning stage
14:          $n \leftarrow n + 1$ 
15:         if  $f = 0$ ; do # D2E phase
16:           Minimize  $\mathcal{L} \leftarrow \mathcal{L}_{\text{enh}} + \mathcal{L}_{\text{uns}}$ 
17:         else # E2D phase
18:           Minimize  $\mathcal{L} \leftarrow \mathcal{L}_{\text{det}_o} + \lambda_1 \mathcal{L}_{\text{det}_e} + \lambda_2 \mathcal{L}_{\text{da}}$ 
19:         if  $\text{mode}(n, n_c) = 0$ ; change  $f$ 
20:   end for

```

confusion within the network. Therefore, the Burn-In stage is split into two distinct phases: the Burn-In-UOD (BID) phase and the Burn-In-UIE (BIE) phase, as depicted in Fig. 4. In the BID phase, images from the original domain $\mathcal{D}_o$, equipped with labeled annotations, are used for UOD tasks. The parameters of the UIE head are frozen during this phase to prevent any interference. These images undergo strong augmentations such as random color jittering, blurring, horizontal flipping, and cropping. On the other hand, the BIE phase uses images from

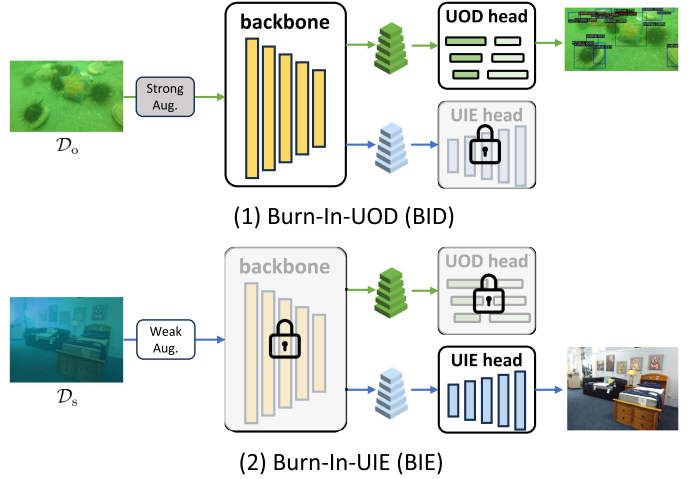


Fig. 4. Our burn-in stage consists of two phases. (1) BID phase and (2) BIE phase.

the synthetic domain $\mathcal{D}_s$ for UIE tasks, freezing the parameters of both the backbone and the UOD head to minimize task conflict. Images in this phase are subjected to weak augmentations, including horizontal flipping and cropping. The network alternates between these phases throughout the Burn-In stage, systematically developing the skills necessary for both UIE and UOD tasks.

2) Mutual-Learning Stage: After the burn-in stage, the network is equipped with the foundational capabilities required for UOD and UIE tasks. To further enhance the network's performance, we introduce the mutual-learning stage to exchange knowledge between domains $d \in \{\mathcal{D}_o, \mathcal{D}_e\}$. As illustrated in Fig. 5, the stage consists of two phases: UOD-to-UIE (D2E) and UIE-to-UOD (E2D) phases. In the D2E phase, the network processes both images from the original domain $\mathcal{D}_o$ and the synthetic domain $\mathcal{D}_s$ to improve the UIE head's ability to handle real-world underwater images. Notably, only the parameters of UOD head are frozen during this phase since images from the

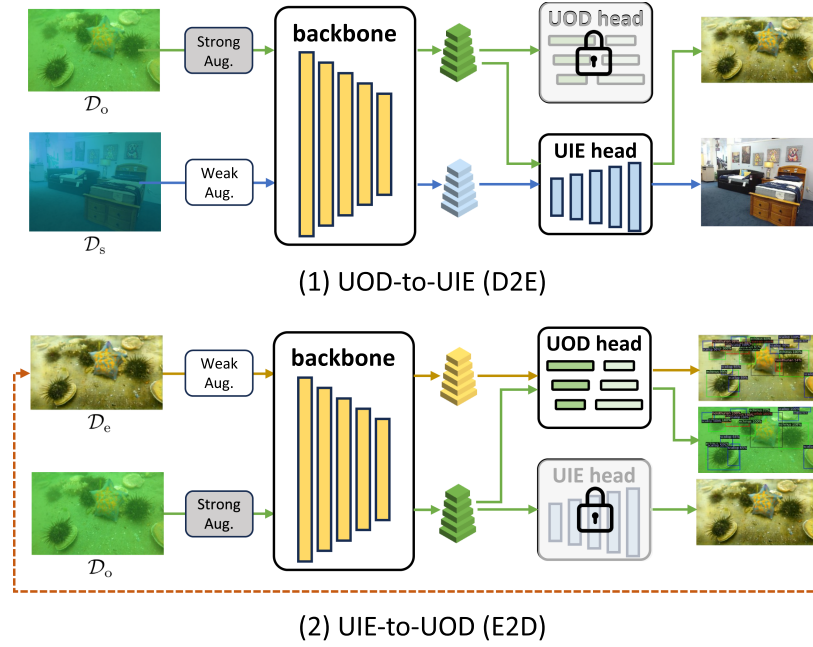


Fig. 5. Our mutual-learning stage is divided into two phases. (1) D2E phase and (2) E2D phase.

original domain $\mathcal{D}_o$ have been incorporated for optimizing the UIE task. Moreover, as there are no clear references for images from $\mathcal{D}_o$, we implement an unsupervised loss $\mathcal{L}_{\text{uns}}$ (as seen in Section III-D) to guide the optimization of the UIE task head. Alternatively, in the E2D phase, enhanced results are generated with frozen UIE head to constitute images in the enhanced domain $\mathcal{D}_e$. These enhanced images, along with those from the original domain $\mathcal{D}_o$, are then used to bolster UOD performance. Furthermore, our proposed domain-adaptation mechanism is employed in this phase to align the encoded features from the original domain $\mathcal{D}_o$ with those from the enhanced domain $\mathcal{D}_e$ and to enhance the distinguishability of less-represented objects across various environments, as detailed in Section III-C.

C. Domain-Adaptation Mechanism

Underwater images commonly exhibit various degraded imaging effects due to different environmental conditions. For example, wavelength-dependent light absorption often results in color shifts in these images, and the backscattering and suspended particles usually contribute to the hazy effect [17]. Although domain adaptation technologies have been extensively studied for object detection in challenging terrestrial environments [19], [38], [41], applying these methods to underwater scenarios remains challenging due to the difficulty in obtaining clear underwater images. To address this issue, we leverage feature maps from images in the enhanced domain $\mathcal{D}_e$ as references to develop a novel domain-adaptation technique for UOD. Our proposed domain-adaptation method encompasses two levels of adaptation: image-level feature alignment (Img-Adap) and object-level feature contrast (Obj-Adap).

1) *Image-Level Feature Alignment*: Due to the lack of clear real-world underwater images, we utilize the enhanced images

from domain $\mathcal{D}_e$ as the clear reference. Therefore, the feature embedding $E(X_o)$ of the input images from the original domain $\mathcal{D}_o$ is aimed to align with the embedding $E(X_e)$ of the images from the enhanced domain $\mathcal{D}_e$. Instead of utilizing adversarial learning for feature alignment [19], [38], [41], which involves extra network components and increases training complexity, our approach simplifies the process by effectively aligning image-level features across domains through minimizing the second-order statistical discrepancy between $E(X_o)$ and $E(X_e)$. The proposed image-level feature alignment loss $\mathcal{L}_{\text{im}}$ is expressed as

$$\mathcal{L}_{\text{im}} = \|\mathcal{C}(E(X_o)) - \mathcal{C}(E(X_e))\|_F^2 \quad (1)$$

where $\mathcal{C}(\cdot)$ represents the feature covariance matrix [42] and $\|\cdot\|_F^2$ denotes the squared Frobenius norm.

2) *Object-Level Feature Contrast*: Ideally, aligning features at the image-level could facilitate the alignment of object-level feature representations [19]. However, this approach may not improve the detection performance for less-represented categories [43]. To address this, we propose Obj-Adap, a domain-adaptation method designed to further mitigate domain shifts for objects within the same category while enhancing feature disparities for objects from different categories. Fig. 6 illustrates that object-level features f_i^d are derived using the RoIAlign operation [44] from image-level features E^d and network predictions p_i^d . Specifically, $f_i^d = \text{RoIAlign}(E^d, p_i^d)$, where $d \in \{\mathcal{D}_o, \mathcal{D}_e\}$ specifies the domain of the input features and predictions, and i denotes the index of the prediction. Subsequently, features f_i^d within the same category c , regardless of their domains, are considered as positive pairs and should be “pulled” closer together. Conversely, features from different categories are recognized as negative pairs and should be “pushed” further apart to increase their separation.

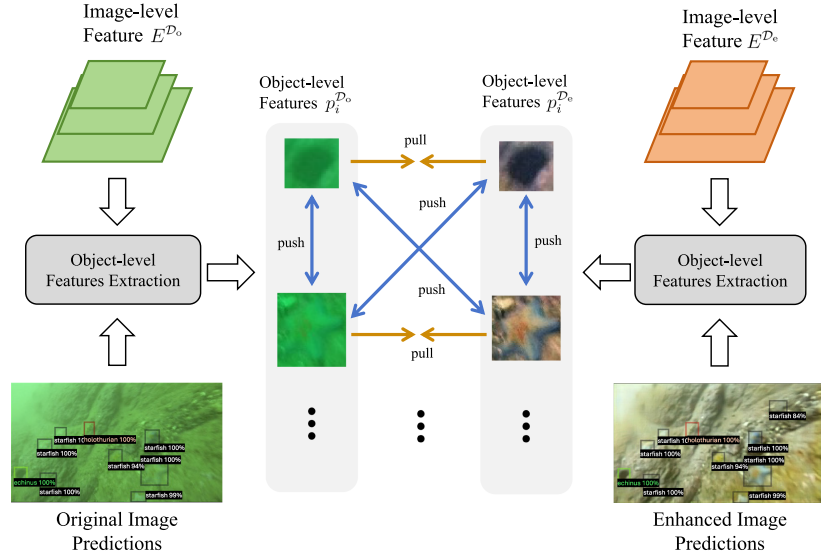
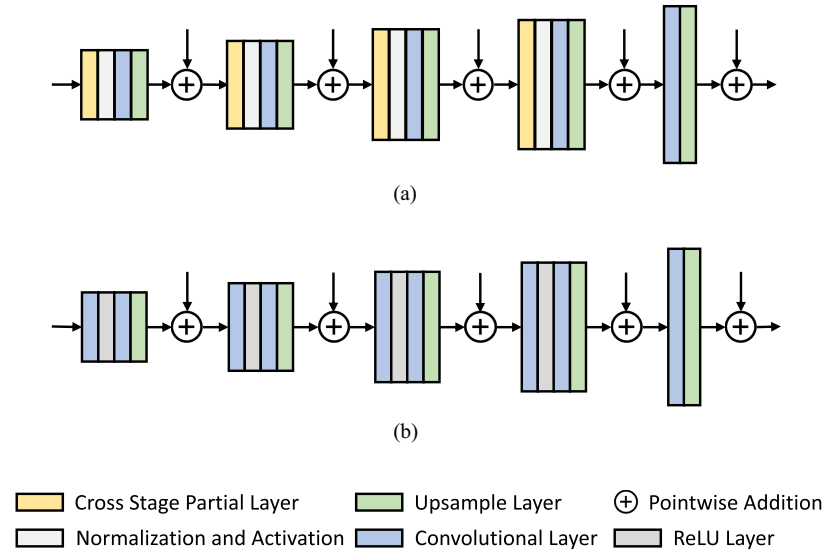


Fig. 6. Illustration of object-level feature contrastive learning.

Fig. 7. Depicted architectures of the UIE heads for (a) EnDet_{YL} and (b) EnDet_{FR}.

Therefore, we propose an object-level contrastive loss $\mathcal{L}_{ob}$, designed to attract positive pairs and repel negative pairs

$$\mathcal{L}_{ob} = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(f_i^d \cdot f_p^d / \tau)}{\sum_{a \in A(i)} \exp(f_i^d \cdot f_a^d / \tau)} \quad (2)$$

where $i \in I \equiv \{1, 2, \dots, N\}$ indexes the filtered predictions in a training batch, $A(i) \equiv I \setminus \{i\}$ represents all predictions except the i th, $P(i) \equiv \{p \in A(i) : c_i^d = c_p^d\}$ denotes the set of indices of all positive pairs with the i th prediction, $|P(i)|$ is its count, and $\tau > 0$ is a positive temperature parameter.

D. Loss Design

Due to the proposed progressive training strategy, the training loss varies across different training stages. Consequently, we will sequentially introduce the losses employed to optimize our framework for each stage.

1) *Burn-In Stage*: In this stage, training is divided into two phases: the BID and BIE phases. During the BID phase, the parameters of the UIE head are fixed, and the optimization is conducted solely for the UOD task. As a result, the training loss $\mathcal{L}$ in the BID phase is equivalent to the detection loss of the original domain images

$$\mathcal{L} = \mathcal{L}_{det_o} \quad (3)$$

where the specific form of the detection loss $\mathcal{L}_{\text{det}_o}$ is determined by the type of detector employed, aligning with the respective loss designs of either one-stage or two-stage detectors [1], [3].

During the BIE phase, the parameters of both the backbone and the UOD head are frozen, while the UIE task head is optimized using images from the synthetic domain. The training loss $\mathcal{L}$ for the BIE phase is then defined as the enhancement loss $\mathcal{L}_{\text{enh}}$ for the UIE head

$$\mathcal{L} = \mathcal{L}_{\text{enh}} = \|I_s - \tilde{I}_s\| \quad (4)$$

where $\|\cdot\|$ denotes the L1 norm, I_s refers to the clear reference in the synthetic domain $\mathcal{D}_s$, and $\tilde{I}_s$ represents the corresponding enhanced results.

2) *Mutual-Learning Stage*: In this stage, the training process is also split into two phases: the D2E and E2D phases. During the D2E phase, images from the original domain $\mathcal{D}_o$ are processed by the UIE task head to enhance its ability to handle real-world underwater images. Since there are no explicit clear references available for images from $\mathcal{D}_o$, an unsupervised loss $\mathcal{L}_{\text{uns}}$ is employed to direct the optimization of the UIE head. Given the assumption that natural images conform to certain fundamental principles, such as the widely recognized “gray-world” assumption [45], we introduce an unsupervised loss $\mathcal{L}_{\text{uns}}$ derived from this principle

$$\mathcal{L}_{\text{uns}} = \frac{1}{3} \sum_c \left(\frac{1}{N} \sum_{i,j} \left(\tilde{I}_o(i,j) - 0.5 \right) \right)^2 \quad (5)$$

where $\tilde{I}_o$ represents the enhanced results from $\mathcal{D}_o$, c is the color channel, and N denotes the number of pixels in $\tilde{X}_o$.

Therefore, in the D2E phase, the training loss is the combination of the enhanced loss $\mathcal{L}_{\text{enh}}$ and the unsupervised loss $\mathcal{L}_{\text{uns}}$

$$\mathcal{L} = \mathcal{L}_{\text{enh}} + \mathcal{L}_{\text{uns}}. \quad (6)$$

In the E2D phase, the enhanced results from the UIE head formulate the images of the enhanced domain $\mathcal{D}_e$ with annotations. These images, together with those from the original domain $\mathcal{D}_o$, are input into the network to improve the performance of the UOD head. The detection losses for images from the enhanced and original domains are denoted as $\mathcal{L}_{\text{det}_e}$ and $\mathcal{L}_{\text{det}_o}$, respectively. Meanwhile, as described in Section III-C, the domain adaptation is also operated in the E2D phase. Consequently, the total training loss in the E2D phase is expressed as

$$\mathcal{L} = \mathcal{L}_{\text{det}_o} + \lambda_1 \mathcal{L}_{\text{det}_e} + \lambda_2 \mathcal{L}_{\text{da}} \quad (7)$$

where $\lambda_{1,2}$ denote the balanced weights of each loss component and $\mathcal{L}_{\text{da}} = \mathcal{L}_{\text{im}} + \mathcal{L}_{\text{ob}}$ represents the domain-adaptation loss.

IV. EXPERIMENTS

A. Data Sets

The synthetic underwater images in $\mathcal{D}_s$ utilized for the UIE task are sourced from SYREA data set [17], including 20 688 training and 1138 testing pairs synthesized from both indoor and outdoor in-air images.

To assess the performance of the UOD, we employ three annotated data sets designed specifically for detection in underwater environments: the DUO data set [21], the RUOD data set [10], and the Brackish data set [46]. The DUO data set includes 6671 training images and 1111 testing images, annotated for categories, such as holothurian (10.64%), echinus (67.12%), scallop (2.67%), and starfish (19.58%). The RUOD data set contains 9800 training images and 4200 testing images, featuring annotations for a wider range of categories including fish (17.50%), turtle (5.44%), jellyfish (3.62%), holothurian (10.10%), starfish (11.06%), diver (8.46%), corals (11.81%), cuttlefish (7.16%), scallop (9.70%), and echinus (15.15%). The Brackish data set includes 9976 training and 1238 testing images, spanning six annotated categories: starfish (22.23%), crab (34.64%), shrimp (1.46%), fish (9.50%), small fish (30.35%), and jellyfish (1.82%).

B. Implementation Details

Our framework has been adapted for compatibility with both one-stage and two-stage detectors. For the one-stage detector configuration, we use YOLOv5 [3] as the baseline network. Our framework maintains the backbone and the UOD head in alignment with YOLOv5’s architecture, while the UIE head is specifically designed based on UNet [47]. Thus, the one-stage version of our EnDet is denoted as EnDet_{YL}. In the case of the two-stage detector, we utilize Faster R-CNN [1] as the baseline. Both the backbone and UOD head are designed to align with Faster R-CNN’s specifications, with a specifically redesigned UIE head. The two-stage version is designated as EnDet_{FR}. The depicted architectures of the UIE heads for EnDet_{YL} and EnDet_{FR} are presented in Fig. 7.

During training, all images are scaled by resizing the longer side to 640 pixels, while preserving their original aspect ratio. The Burn-In epoch $E_b = 200$, and the training continues up to a maximum epoch $E_{\text{max}} = 300$. The phase change step n_c , used in both the burn-in and mutual-learning stages, is empirically set to 200. This value was selected based on experimental analysis, where values ranging from 50 to 300 were tested. Our findings indicated that $n_c = 200$ provided the optimal balance between training stability and task integration effectiveness. Smaller steps led to unstable convergence due to frequent alternation, whereas larger steps hindered the timely mutual benefits of joint learning. In addition, an early stop strategy [48] is employed to prevent overfitting during training. The base learning rate is initially set at 0.001 and is scheduled to decay by a factor of 0.1 after 200 epochs. The balanced weights, λ_1 and λ_2 , are set as 1.0 and 0.1, respectively. The batch size for UOD input data is 16, while the batch size of UIE data are set at 8. The training processes are conducted using PyTorch on two NVIDIA RTX 3090 GPUs, employing the SGD optimizer.

C. Evaluation Metrics and Compared Methods

The performance of object detectors for a specific category is evaluated using Average Precision (AP). For overall detection performance, we use mean average precision (mAP) and mAP_{0.5} as evaluation metrics. mAP measures the AP across multiple

TABLE I
DETAILS OF THE COMPARED UOD METHODS

Method	Type	Backbone	Architecture Design	Training Strategy
YOLOv5 [3] (baseline)	One-stage	DarkNet53	Direct	Origin-then-UOD
FunieGAN [49]+YOLOv5 [3]	One-stage	DarkNet53	Sequential	UIE-then-UOD
SyreaNet [17]+YOLOv5 [3]	One-stage	DarkNet53	Sequential	UIE-then-UOD
GLNet [32]+YOLOv5 [3]	One-stage	DarkNet53	Sequential	UIE-then-UOD
TUDA [18]+YOLOv5 [3]	One-stage	DarkNet53	Sequential	UIE-then-UOD
CSAM [6]	One-stage	DarkNet53	Sequential	Joint-UIE-UOD
DE-YOLO [50]	One-stage	DarkNet53	Sequential	Joint-UIE-UOD
IA-YOLO [20]	One-stage	DarkNet53	Sequential	Joint-UIE-UOD
UnitModule [8]	One-stage	DarkNet53	Sequential	Joint-UIE-UOD
IEOD [12]	One-stage	DarkNet53	Parallel	Joint-UIE-UOD
IFG-UOD [15]	One-stage	DarkNet53	Parallel	Joint-UIE-UOD
EnDet_{YL}(Ours)	One-stage	DarkNet53	Multi-Task	Joint-UIE-UOD
FasterRCNN [1] (baseline)	Two-stage	ResNet50	Direct	Origin-then-UOD
FunieGAN [49]+FasterRCNN [1]	Two-stage	ResNet50	Sequential	UIE-then-UOD
SyreaNet [17]+FasterRCNN [1]	Two-stage	ResNet50	Sequential	UIE-then-UOD
GLNet [32]+FasterRCNN [1]	Two-stage	ResNet50	Sequential	UIE-then-UOD
TUDA [18]+FasterRCNN [1]	Two-stage	ResNet50	Sequential	UIE-then-UOD
RoIMix [4]	Two-stage	ResNet50	Direct	Origin-then-UOD
BoostingRCNN [5]	Two-stage	ResNet50	Direct	Origin-then-UOD
DJL-Net [13]	Two-stage	ResNet50	Parallel	Joint-UIE-UOD
JADSNet [14]	Two-stage	ResNet50	Multi-Task	Joint-UIE-UOD
EnDet_{FR}(Ours)	Two-stage	ResNet50	Multi-Task	Joint-UIE-UOD

intersection-over-union (IoU) thresholds, ranging from 0.5 to 0.95 in increments of 0.05, while $mAP_{0.5}$ evaluates precision specifically at an IoU threshold of 0.5.

We conduct comprehensive experiments on the DUO [21], RUOD [10], and Brackish [46] data sets, comparing a wide range of SOTA methods, as summarized in Table I. For a fair comparison, all one-stage detectors are adapted to use YOLOv5's backbone, while all two-stage detectors are modified to use FasterRCNN's backbone. Furthermore, all methods are retrained on the training sets of the DUO, RUOD, and Brackish data sets. For UIE-then-UOD methods, underwater images are first processed by the UIE method, and the enhanced images are then used for subsequent UOD training and testing.

D. Experiments on Detection Performance

Table II presents the quantitative results of various methods on the DUO [21] test set. Sequential methods using a UIE-then-UOD training strategy consistently show a decline in detection performance compared to their baseline models. This finding aligns with prior studies [9], [10], [11], which highlight that while UIE methods may improve image perceptual quality, they do not necessarily enhance downstream vision task performance. In contrast, sequential methods employing a Joint-UIE-UOD training strategy demonstrate improved UOD performance. For instance, DE-YOLO [50], although based on a sequential architecture, achieves a +1.51% mAP improvement over YOLOv5. This improvement is attributed to its joint training strategy, which directly optimizes image enhancement to support the UOD task.

Methods employing parallel and multitask architectures show only marginal improvements over their baseline models, likely due to their reliance on existing UIE methods, which may limit UOD performance. For example, IEOD [12] uses a consistency loss to align UOD features with those enhanced by UIEC2Net [29], while JADSNet uses enhanced results from UIE methods [30] as references. In comparison, our one-stage model, EnDet_{YL}, and two-stage model, EnDet_{FR}, significantly outperform their baselines, achieving mAP improvements of $\pm 2.63\%$ and $\pm 3.71\%$, respectively. Notably, in the scallop category, EnDet_{YL}/EnDet_{FR} show remarkable gains of $\pm 3.95\%/\pm 3.14\%$ mAP, largely attributed to the proposed Obj-Adap mechanism discussed in Sections IV-E and IV-H3.

The visual comparison on the DUO [21] test set is shown in Fig. 8. While TUDA+YOLOv5 produces visually appealing images, these enhancements do not necessarily improve UOD performance, as they are not aligned with detection requirements. DE-YOLO [50], on the other hand, enhances UOD results compared to the baseline YOLOv5 [3], but its improvements are primarily focused on detection algorithms and do not enhance human visibility. In contrast, the proposed EnDet_{YL} improves both image quality and detection performance, demonstrating its effectiveness in simultaneously addressing visual enhancement and UOD objectives.

The quantitative results on the RUOD [10] and Brackish [46] test sets are summarized in Table III. Similar to the trends observed in the DUO data set, UIE-then-UOD methods that use UIE as a preprocessing step show a decline in performance compared to their baselines, while other Joint-UIE-UOD methods provide marginal improvements. In contrast, our proposed

TABLE II
QUANTITATIVE COMPARISON OF DETECTION RESULTS FOR VARIOUS METHODS ON DUO [21] TEST SET

Method	mAP (%)	mAP _{0.5} (%)	mAP _{small}	mAP _{medium}	mAP _{large}	AP _{holothurian}	AP _{scallop}
YOLOv5 [3] (baseline)	57.51	83.23	42.39	67.63	69.34	56.71	43.18
FunieGAN [49]+YOLOv5 [3]	55.28 (-2.23)	81.86	41.57	64.78	65.43	56.01	40.91
SyreaNet [17]+YOLOv5 [3]	57.02 (-0.49)	82.81	42.62	66.48	69.51	57.05	44.01
GLNet [32]+YOLOv5 [3]	57.09 (-0.42)	83.01	42.37	66.87	69.78	57.23	43.01
TUDA [18]+YOLOv5 [3]	54.61 (-2.90)	81.01	39.48	64.58	63.22	55.43	41.62
CSAM [6]	57.92 (+0.41)	82.98	42.85	68.19	70.01	57.55	43.02
DE-YOLO [50]	<u>59.02</u> (+1.51)	84.13	44.03	<u>69.12</u>	70.22	<u>59.49</u>	<u>45.81</u>
IA-YOLO [20]	58.21 (+0.70)	83.45	42.33	67.85	68.62	57.94	43.11
UnitModule [8]	58.44 (+0.93)	83.07	44.58	68.77	70.34	58.22	43.80
IEOD [12]	57.68 (+0.17)	82.77	42.32	67.81	69.24	57.01	43.08
IFG-UOD [15]	58.18 (+0.67)	81.54	43.55	67.32	70.41	57.81	43.10
EnDet_{YL} (Ours)	60.14 (+2.63)	<u>83.67</u>	46.91	69.53	74.73	61.63	47.13
FasterRCNN [1] (baseline)	55.22	78.53	40.33	65.87	68.02	56.43	41.51
FunieGAN [49]+FasterRCNN [1]	52.16 (-3.06)	78.42	39.54	63.89	62.17	55.81	40.37
SyreaNet [17]+FasterRCNN [1]	54.97 (-0.25)	79.08	40.35	64.76	64.24	55.92	41.02
GLNet [32]+FasterRCNN [1]	54.92 (-0.30)	79.12	40.08	65.17	65.28	56.10	40.96
TUDA [18]+FasterRCNN [1]	53.17 (-2.05)	78.05	39.54	63.67	63.85	55.04	41.88
RoIMix [4]	56.81 (+1.59)	81.35	40.29	66.31	67.54	55.38	42.72
BoostingRCNN [5]	55.68 (+0.46)	80.27	41.32	66.22	67.01	55.29	41.79
DJL-Net [13]	57.02 (+1.80)	80.33	41.28	66.95	69.10	56.85	43.02
JADSNet [14]	56.54 (+1.32)	80.08	41.55	66.64	69.31	56.84	42.33
EnDet_{FR} (Ours)	58.93 (+3.71)	82.94	43.78	68.01	<u>70.51</u>	58.36	44.65

Note: **Bold** denotes the highest score and underline indicates the second highest. **Red** signifies mAP decrease while **blue** represents increase.

TABLE III
QUANTITATIVE COMPARISON OF DETECTION RESULTS FOR VARIOUS METHODS ON RUOD [10] AND BRACKISH [46] TEST SETS

Method	Training Strategy	RUOD [10]		Brackish [46]	
		mAP (%)	mAP _{0.5} (%)	mAP (%)	mAP _{0.5} (%)
YOLOv5 [3] (baseline)	Origin-then-UOD	53.46	83.41	72.87	96.33
FunieGAN [49]+YOLOv5 [3]	UIE-then-UOD	52.83 (-0.63)	82.78	71.47 (-1.40)	96.14
SyreaNet [17]+YOLOv5 [3]	UIE-then-UOD	53.02 (-0.44)	82.81	72.42 (-0.45)	96.27
GLNet [32]+YOLOv5 [3]	UIE-then-UOD	53.27 (-0.19)	83.01	72.55 (-0.32)	96.41
TUDA [18]+YOLOv5 [3]	UIE-then-UOD	52.61 (-0.85)	82.59	70.94 (-1.93)	95.97
CSAM [6]	Joint-UIE-UOD	54.38 (+0.92)	83.43	73.45 (+0.58)	96.53
DE-YOLO [50]	Joint-UIE-UOD	<u>55.89</u> (+2.43)	83.79	73.86 (+0.99)	97.02
IA-YOLO [20]	Joint-UIE-UOD	54.72 (+1.26)	83.48	73.22 (+0.35)	96.02
UnitModule [8]	Joint-UIE-UOD	55.75 (+2.29)	<u>84.35</u>	<u>74.17</u> (+1.30)	97.14
IEOD [12]	Joint-UIE-UOD	53.99 (+0.53)	83.51	73.29 (+0.42)	97.07
IFG-UOD [15]	Joint-UIE-UOD	54.07 (+0.61)	83.68	73.56 (+0.69)	96.53
EnDet_{YL} (Ours)	Joint-UIE-UOD	58.06 (+4.60)	85.62	75.13 (+2.26)	98.04
FasterRCNN [1] (baseline)	Origin-then-UOD	52.71	83.17	69.81	95.82
FunieGAN [49]+FasterRCNN [1]	UIE-then-UOD	51.84 (-0.87)	82.61	67.15 (-2.66)	94.77
SyreaNet [17]+FasterRCNN [1]	UIE-then-UOD	52.32 (-0.39)	82.91	68.45 (-1.36)	95.50
GLNet [32]+FasterRCNN [1]	UIE-then-UOD	52.51 (-0.20)	83.08	68.94 (-0.87)	95.07
TUDA [18]+FasterRCNN [1]	UIE-then-UOD	51.44 (-1.27)	82.56	66.95 (-2.86)	94.31
RoIMix [4]	Origin-then-UOD	53.17 (+0.46)	83.54	70.22 (+0.41)	96.10
BoostingRCNN [5]	Origin-then-UOD	53.02 (+0.31)	83.29	70.14 (+0.33)	95.75
DJL-Net [13]	Joint-UIE-UOD	54.81 +2.10	83.50	72.10 (+2.29)	96.71
JADSNet [14]	Joint-UIE-UOD	54.96 (+2.25)	83.65	71.38 (+1.57)	96.05
EnDet_{FR} (Ours)	Joint-UIE-UOD	55.76 (+3.05)	84.05	72.52 (+2.71)	<u>97.15</u>

Notes: **Bold** denotes the highest score and underline indicates the second highest. **Red** signifies mAP decrease while **blue** represents increase.

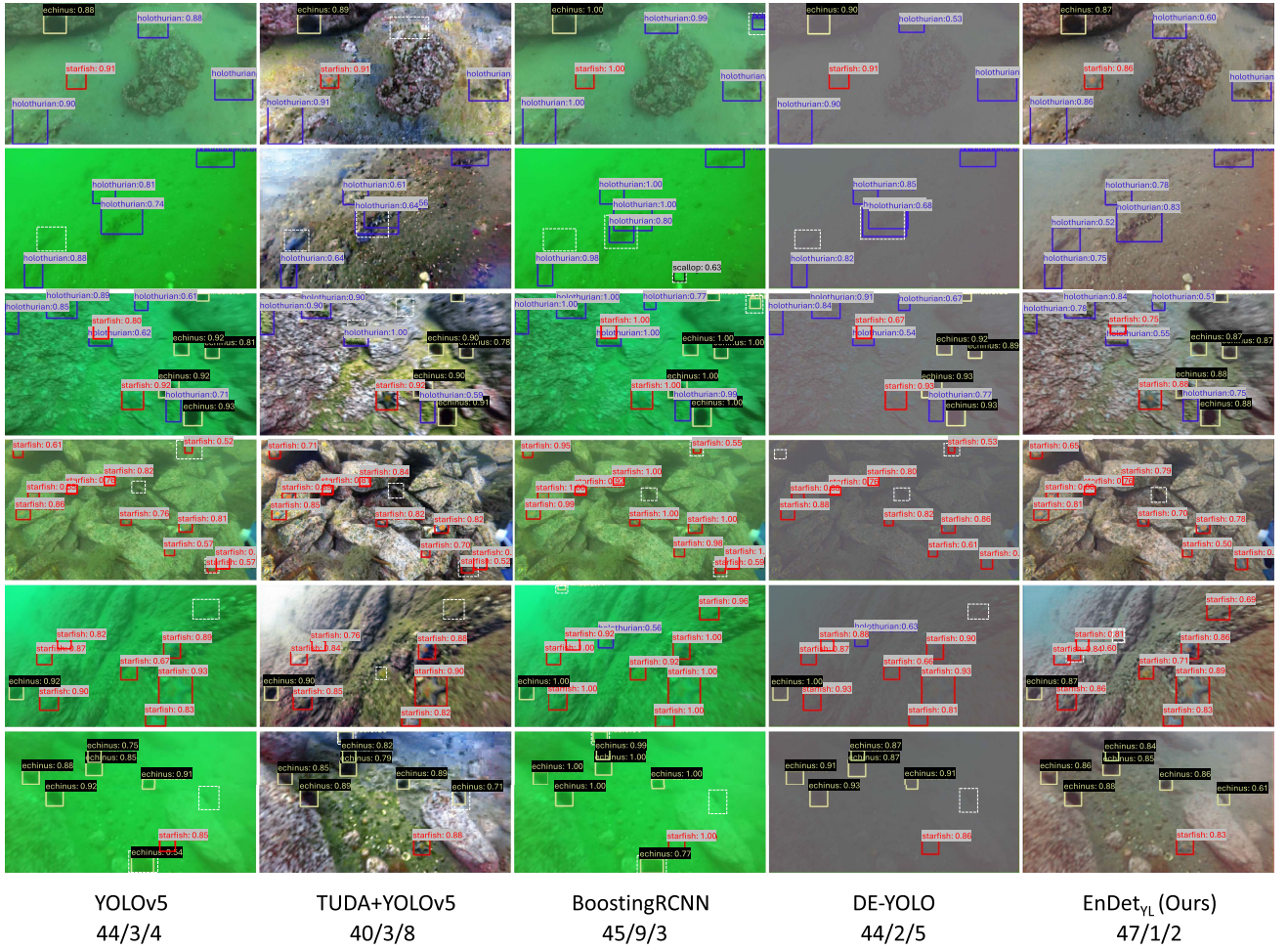


Fig. 8. Detection results on the DUO [21] test set for various UOD methods. Colored rectangles indicate true positives (TP), while white dotted rectangles represent false positives (FP) or false negatives (FN). The numbers below each method indicate the counts of TP/FP/FN. Our proposed method achieves the best performance by detecting the highest number of TPs and the fewest FPs and FNs. Zoom in for a better view.

methods significantly outperform their baselines. Visual results in Fig. 9 further demonstrate the effectiveness of our approach, highlighting its ability to enhance both image quality and detection performance.

E. Experiments on Domain Adaptation

To evaluate domain adaptation capabilities, we use WCT2 [40] to synthesize five new environments (bluish, greenish, yellowish, turbid, and darkish) by transferring styles from real-world underwater images to the DUO [21] test set, as shown in Fig. 10. As reported in Table IV, all methods experience significant performance drops on these domains due to the large domain discrepancies. Fig. 11(a) visually highlights the substantial feature distribution gaps between the original domain and new environments, such as turbid and darkish, which lead to marked performance degradation. In contrast, our proposed method not only achieves superior performance on the original DUO [21] test set but also significantly outperforms other UOD methods on the synthesized domains. For example, in yellowish and darkish environments, EnDet_{FR} surpasses the baseline by over $\pm 15\%$

mAP, largely owing to the proposed Img-Adap mechanism. As illustrated by the qualitative results in Fig. 11(b), the Img-Adap mechanism effectively aligns image-level feature distributions across domains, reducing domain shifts and improving model adaptability.

In addition, we provide qualitative analysis of the Obj-Adap mechanism by visualizing object-level feature distributions before and after adaptation. As shown in Fig. 12, our Obj-Adap mechanism improves intracategory compactness and intercategory separation, making category-specific features more distinguishable. This visualization further supports the effectiveness of Obj-Adap in enhancing detection robustness across domains. A quantitative analysis of Obj-Adap's influence is provided in Section IV-H3.

F. Experiments on Image Enhancement

We compare our method with other SOTA UIE methods on the DUO [21] data set, including GLNet [32], SyreaNet [17], FunieGAN [49], and TUDA [9]. As shown in Fig. 13, while these methods generally produce visually appealing results,

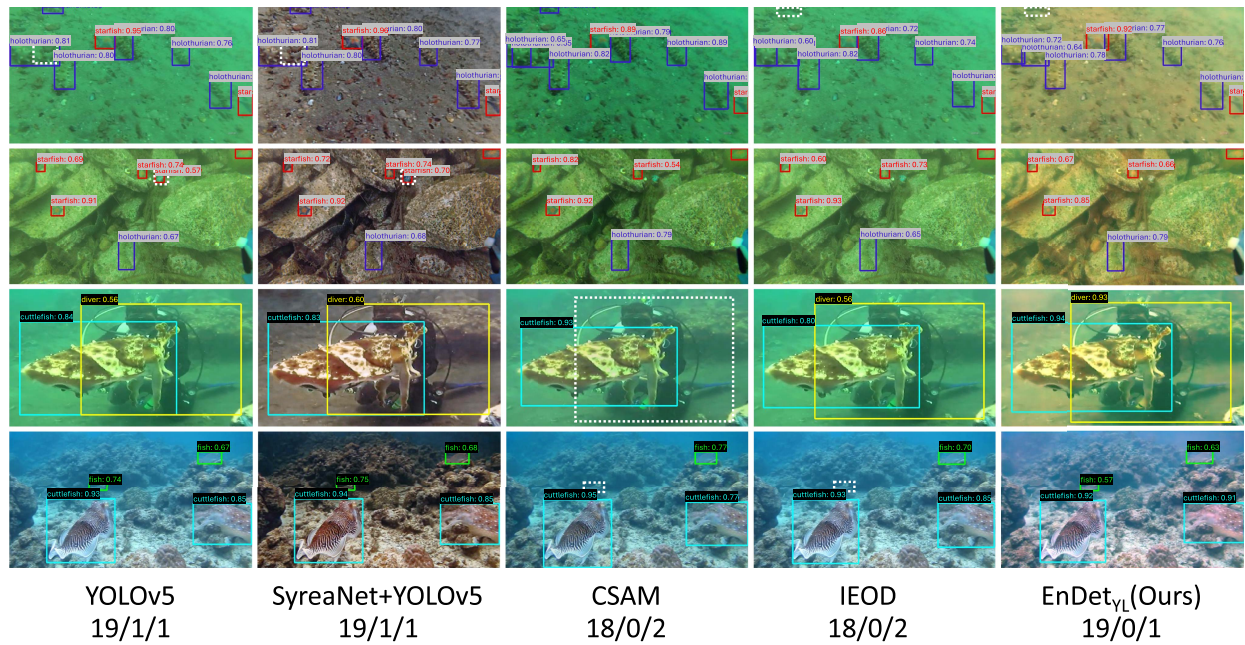


Fig. 9. Detection results on the RUOD [10] test set for various UOD methods. Colored rectangles indicate TP, while white dotted rectangles represent FP or PN. The numbers below each method indicate the counts of TP/FP/FN. Zoom in for a better view.

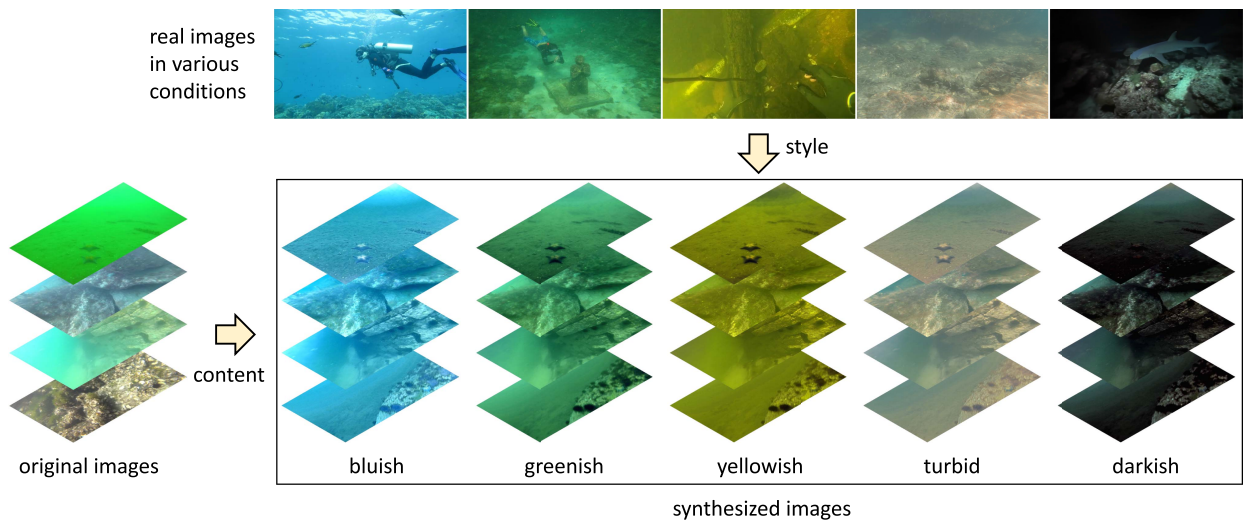


Fig. 10. Synthesized results using WCT2 [40]: the first row shows real underwater images under various conditions, providing the style. The first column displays original images from the DUO [21] data set, serving as the content. The remaining images are synthesized results that retain the content of the original images while adopting the style of the corresponding underwater conditions.

they often introduce undesired artifacts. For example, in Row 3, TUDA [9] generates significant artifacts in the background, potentially confusing underwater object detectors. Similarly, in Row 4, both GLNet [32] and TUDA [9] alter the greenish water to a bluish tone, which could negatively impact downstream vision tasks. In contrast, our method introduces fewer artifacts and produces more consistent enhancements, making it better suited for underwater applications.

To quantitatively evaluate enhancement performance, we calculate several nonreference image quality assessment metrics specifically designed for underwater images, including

UIQM [51], UCIQE [22], TMix [52], and URanker [53]. The results, shown in Table V, indicate that while our method substantially improves the quality of degraded underwater images, it does not achieve the highest scores compared to some UIE methods solely optimized for visual quality, such as TUDA [9] and GLNet [32]. This is primarily because our framework is designed to optimize features relevant for UOD rather than focusing exclusively on perceptual quality. Furthermore, current nonreference metrics do not always reflect the suitability of enhanced images for downstream tasks. As shown in Fig. 13 and Table V, although TUDA [9] attains high metric scores,

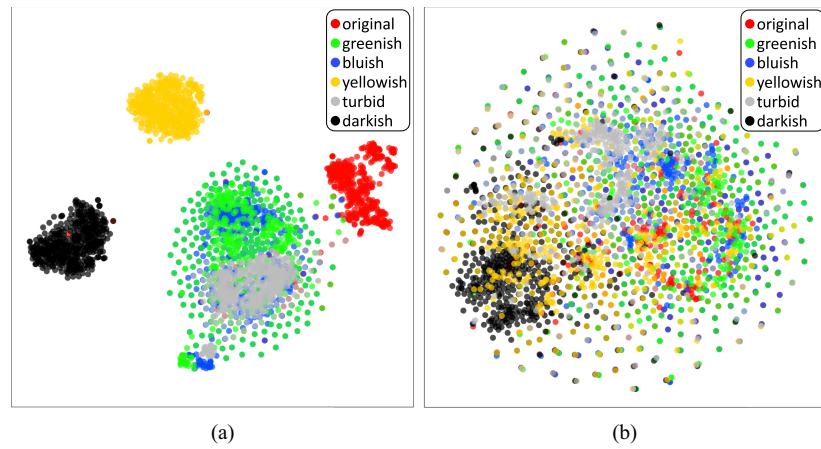


Fig. 11. t-SNE visualization of image-level features across different underwater environments before and after applying the Img-Adap mechanism. (a) Before adaptation, significant domain gaps exist between the original and synthesized environments. (b) After adaptation, the Img-Adap mechanism effectively aligns feature distributions across domains, reducing domain discrepancies and demonstrating improved adaptability. (a) Before Img-Adap (b) After Img-Adap.

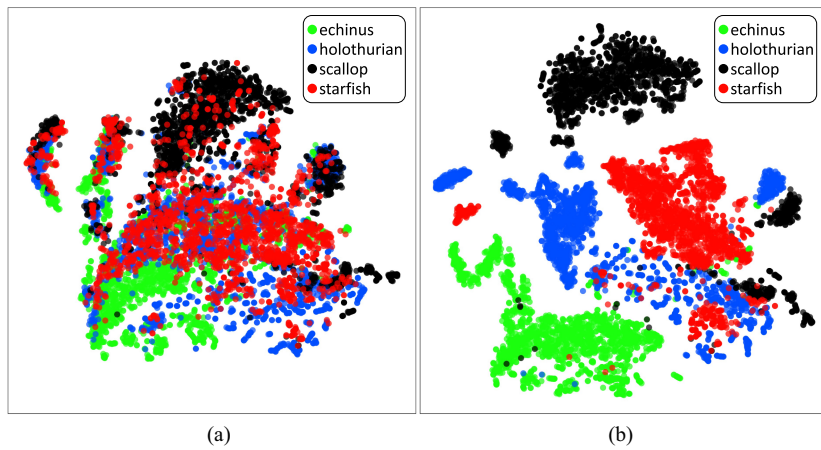


Fig. 12. t-SNE visualization of object-level features on the DUO test set before and after applying the Obj-Adap mechanism. (a) Before adaptation, features of different categories show less distinction and overlap. (b) After adaptation, the Obj-Adap mechanism enhances intra-category compactness and intercategory separation, making category-specific features more distinguishable. (a) Before Obj-Adap (b) After Obj-Adap.

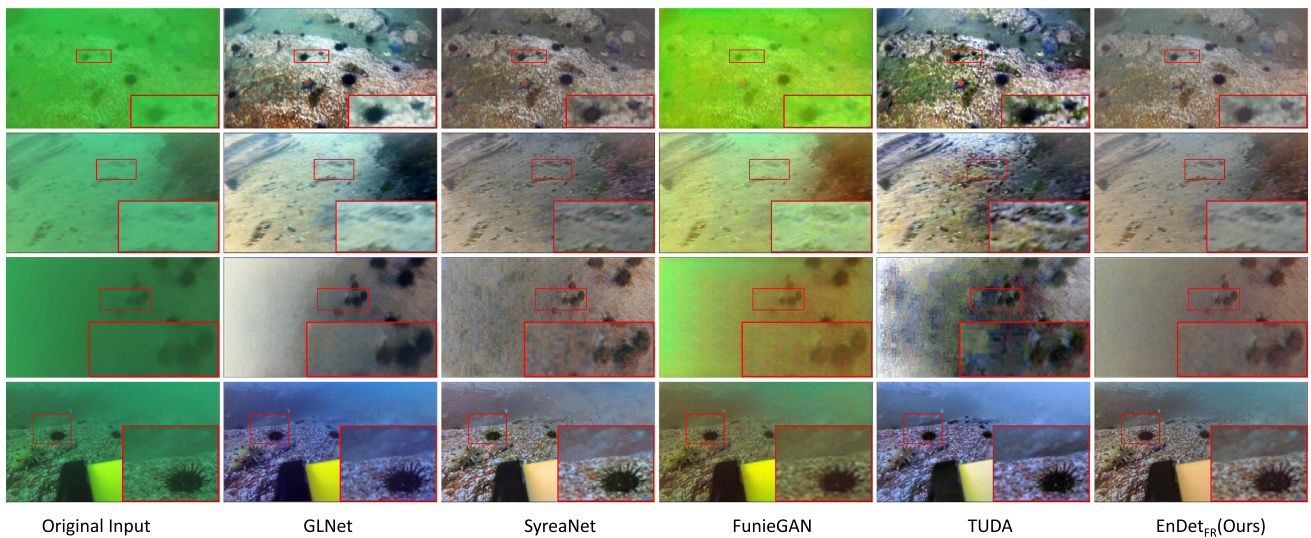


Fig. 13. Visual comparison of various UIE results on DUO [21] data set. Zoomed-in inset images clearly illustrate the differences in object-level details and artifact presence among compared methods. Zoom in further for a better view.

TABLE IV
MAP (%) SCORES OF VARIOUS METHODS ON DUO [21] TEST SET AND UNSEEN ENVIRONMENTS

Method	original	bluish	greenish	yellowish	turbid	darkish
YOLOv5 [3] (baseline)	57.51	30.41	46.73	29.64	20.50	22.35
CSAM [6]	57.92	33.61	48.98	31.04	25.18	23.21
DE-YOLO [50]	<u>59.02</u>	34.21	47.41	<u>32.87</u>	26.28	24.55
IA-YOLO [20]	58.21	33.81	48.52	31.21	24.56	25.18
UnitModule [8]	58.44	30.31	47.55	30.50	23.58	26.31
IEOD [12]	57.68	35.01	47.58	31.15	26.41	24.79
IFG-UOD [15]	58.18	32.88	49.12	30.76	25.88	25.31
EnDet _{YL} (Ours)	60.14	41.31	51.93	35.86	29.29	30.46
FasterRCNN [1] (baseline)	55.22	22.55	37.31	16.24	21.25	12.79
RoIMix [4]	56.81	27.52	42.55	25.72	24.80	20.95
BoostingRCNN [5]	55.68	29.76	40.17	24.28	25.33	22.05
DJL-Net [13]	57.02	28.53	46.21	29.58	25.45	25.90
JADSNet [14]	56.54	25.88	45.22	28.34	24.52	26.12
EnDet _{FR} (Ours)	58.93	<u>35.87</u>	<u>49.21</u>	32.34	<u>27.13</u>	<u>28.53</u>

Notes: **Bold** denotes the highest score and underline indicates the second highest.

TABLE V
QUANTITATIVE COMPARISON OF ENHANCEMENT PERFORMANCE ON DUO [21] TEST SET FOR OUR PROPOSED METHOD AND OTHER UIE METHODS

Metrics	Original	GLNet [32]	SyreaNet [17]	FunieGAN [49]	TUDA [9]	EnDet _{YL}	EnDet _{FR}
UIQM [51]	0.421	1.143	1.334	1.184	1.416	1.206	1.258
UCIQE [22]	0.420	0.584	0.494	0.487	0.571	0.527	0.532
TMix [52]	0.241	0.842	0.646	0.230	0.749	0.653	0.719
URanker [53]	1.216	1.673	1.598	1.535	1.833	1.528	1.587

it often introduces noticeable artifacts. These visual artifacts are not adequately evaluated by existing metrics, which explains the discrepancy between quantitative scores and qualitative assessment. Nevertheless, our method achieves a more balanced performance across both detection and enhancement tasks, delivering the best UOD performance while also ensuring effective and visually reliable image enhancement. In contrast, other UIE-then-UOD methods, although effective in improving certain perceptual metrics, tend to exhibit a decline in detection performance compared to their baselines.

G. Framework Modes and Efficiency Analysis

We propose three operational modes for our framework in real-world testing: detection mode, enhance mode, and dual mode, providing flexibility in operation. In detection or enhance mode, the framework performs a single task, either UOD or UIE, minimizing computational overhead. In contrast, dual mode enables simultaneous execution of UIE and UOD tasks, allowing researchers to visualize detection results alongside enhanced underwater images, providing a more comprehensive understanding of the detections.

An efficiency analysis was conducted to assess the practicality of the proposed method for real-world applications. As shown in Table VI, in detection mode, both EnDet_{YL} and EnDet_{FR} achieve

inference times comparable to their respective baseline models. Notably, in dual mode, EnDet_{YL} significantly outperforms other Joint-UIE-UOD and UIE-then-UOD methods, achieving an inference latency of 11.32 ms and a processing speed of 88.34 frames per second (fps), demonstrating its strong potential for real-world deployment. Furthermore, Fig. 2 illustrates the inference speed of each UOD method through the radius of the plotted dots, further highlighting the superiority of the proposed method.

H. Ablation Study

1) *Effectiveness of Burn-In Stage*: The impact of the burn-in stage is shown in Table VII. The Ab₁ model bypasses the burn-in stage, entering directly into the mutual-learning stage, which results in a significant decrease of over -3% mAP compared to the full model EnDet_{YL}, highlighting the importance of establishing foundational UOD and UIE capabilities before pursuing further improvements. The Ab₂ model skips the BIE phase, focusing entirely on UOD training during the burn-in stage, whereas the Ab₃ model omits the BID phase, concentrating solely on optimizing the UIE task head. Notably, the absence of the BID phase leads to a more substantial performance decline, underscoring the critical role of developing initial detection capability.

TABLE VI
EFFICIENCY ANALYSIS (INPUT SIZE 672×384 , ON A SINGLE NVIDIA RTX 3090 GPU)

Methods	Latency ↓	FPS ↑	GFLOPs ↓	Params ↓
YOLOv5 [3]	9.40 ms	105.93	5.01	7.03 M
FasterRCNN [1]	14.40 ms	69.48	60.11	41.36 M
SyreaNet [17]+YOLOv5 [3]	62.60 ms	15.97	329.3	36.09 M
GLNet [32]+YOLOv5 [3]	86.34 ms	11.58	75.39	66.18 M
BoostingRCNN [5]	31.40 ms	31.88	44.39	44.25 M
DE-YOLO [50]	45.20 ms	22.10	10.24	7.05 M
IA-YOLO [20]	46.50 ms	21.53	10.27	8.03 M
IEOD [12]	36.50 ms	27.38	<u>5.02</u>	<u>7.04</u> M
UnitModule [8]	14.30 ms	69.93	8.75	7.06 M
DJL-Net [13]	76.80 ms	13.02	60.69	76.11 M
EnDet _{YL} (Det. Mode)	9.40 ms	105.93	5.01	7.03 M
EnDet _{YL} (Enh. Mode)	<u>10.30</u> ms	<u>97.09</u>	5.31	<u>7.04</u> M
EnDet _{YL} (Dual Mode)	11.32 ms	88.34	6.04	10.54 M
EnDet _{FR} (Det. Mode)	14.40 ms	69.48	60.11	41.36 M
EnDet _{FR} (Enh. Mode)	16.38 ms	61.05	55.62	56.71 M
EnDet _{FR} (Dual Mode)	17.57 ms	56.92	40.58	77.39 M

Notes: **Bold** denotes the best result and underline indicates the second best.

TABLE VII
ABLATION STUDIES ON THE EFFECTIVENESS OF BURN-IN STAGE ACROSS VARIOUS DATA SETS

Ablations	Ab ₁	Ab ₂	Ab ₃	EnDet _{YL}
BID		✓		✓
BIE			✓	✓
mAP (DUO [21])	56.36 (-3.78)	58.22 (-1.92)	56.74 (-3.40)	60.14
mAP (RUOD [10])	54.58 (-3.48)	56.35 (-1.71)	55.10 (-2.96)	58.06
mAP (Brackish [46])	72.05 (-3.08)	73.98 (-1.15)	72.84 (-2.29)	75.13

Notes: **Red** indicates a decrease in mAP (%) compared to the full model

TABLE VIII
ABLATION STUDIES ON THE EFFECTIVENESS OF MUTUAL-LEARNING STAGE ACROSS VARIOUS DATA SETS

Ablations	Ab ₄	Ab ₅	Ab ₆	EnDet _{YL}
D2E		✓		✓
E2D			✓	✓
mAP (DUO [21])	57.41 (-2.73)	56.55 (-3.59)	57.62 (-2.52)	60.14
mAP (RUOD [10])	54.85 (-3.21)	52.40 (-5.66)	55.10 (-2.96)	58.06
mAP (Brackish [46])	72.64 (-2.49)	71.95 (-3.18)	73.48 (-1.65)	75.13

Notes: **Red** indicates a decrease in mAP (%) compared to the full model.

2) *Effectiveness of Mutual-Learning Stage*: The significance of the mutual-learning stage is demonstrated in Table VIII. The Ab₄ model omits the mutual-learning stage, relying solely on the burn-in stage for training. This approach results in a dramatic decline in mAP across all testing sets, highlighting the importance of knowledge exchange between the proposed domains $\mathcal{D}_o$ and $\mathcal{D}_e$ during the mutual-learning stage. The Ab₅ model exhibits a more severe performance drop than the Ab₄ model, attributed to the optimization of both the backbone and the UIE task head during the D2E phase, which can skew network parameters

towards UIE at the expense of UOD effectiveness. In addition, the Ab₆ model, which excludes the D2E phase, also experiences a significant performance decrease, suggesting the importance of enhancing the UIE task head's capabilities during this stage.

3) *Effectiveness of Domain-Adaptation*: The effectiveness of our proposed domain-adaptation mechanism can be seen in Table IX. The Ab₇ model, lacking any domain-adaptation mechanism, experiences only a minor decline in overall mAP compared to the full model. However, more significant decreases are observed in the less-represented scallop category and in

TABLE IX
ABLATION STUDIES FOR THE EFFECTIVENESS OF THE PROPOSED DOMAIN-ADAPTATION MECHANISM ON THE DUO [21] TEST SET

Ablations	Ab ₇	Ab ₈	Ab ₉	EnDet _{YL}
Img-Adap		✓		✓
Obj-Adap			✓	✓
mAP	59.80 (-0.34)	59.92 (-0.22)	59.86 -0.28	60.14
mAP (scallop)	43.25 (-3.88)	43.82 (-3.31)	46.36 (-0.77)	47.13
mAP (darkish)	26.93 (-3.53)	29.48 (-0.98)	27.28 (-3.18)	30.46

Notes: Red indicates a decrease in mAP (%) compared to the full model.

TABLE X
DETECTION PERFORMANCE (MAP AND MAP_{0.5}) OF DIFFERENT METHODS ON THE DUO AND RUOD DATA SETS WITH DIVERSE DETECTOR ARCHITECTURES

Method	DUO [21]		RUOD [10]	
	mAP (%)	mAP _{0.5} (%)	mAP (%)	mAP _{0.5} (%)
CEH-YOLO [57]	63.22	82.87	61.24	<u>85.10</u>
FBDPN [58]	61.55	83.49	61.08	84.29
PE-Transformer [59]	63.55	84.00	62.19	85.07
YOLOv8 [54]	62.75	83.41	60.47	84.53
YOLOv10 [55]	62.31	83.62	60.27	84.85
RT-DETR [56]	63.08	<u>84.10</u>	60.95	84.49
EnDet _{YOLOv8}	64.20	84.25	62.75	85.22
EnDet _{YOLOv10}	63.26	84.03	63.88	85.04
EnDet _{RT-DETR}	<u>63.95</u>	83.94	<u>62.94</u>	84.92

Notes: Bold denotes the highest score and underline indicates the second highest.

TABLE XI
DETECTION RESULTS WITH DIFFERENT $\mathcal{D}_s$ TRAINING IMAGES

$\mathcal{D}_s$ Images	LNRUD [60]	UWCNN [61]	SYREA [17] (ours)	YOLOv5 [3]
mAP (DUO)	58.84 (+1.33)	57.26 (-0.25)	60.14 (+2.63)	57.51
mAP (RUOD)	56.31 (+2.85)	52.17 (-1.29)	58.06 (+4.60)	53.46
mAP (Brackish)	73.69 (+0.82)	72.15 (-0.72)	75.13 (+2.26)	72.87

Notes: Red indicates a decrease in mAP compared to the full model.

new darkish environment, emphasizing the importance of our domain-adaptation mechanism. Specifically, the Ab₈ model, which omits the Obj-Adap mechanism, suffers a substantial decline in mAP for the less-represented scallop category. In contrast, the Ab₉ model, excluding the Img-Adap mechanism, exhibits a significant performance drop in mAP for new darkish environments. These results underscore the critical role of our domain-adaptation mechanisms in maintaining high detection performance across varied conditions.

I. Validation on Diverse Detectors

To further evaluate the robustness and generalization capability of EnDet, we conduct additional experiments on the DUO [21] and RUOD [10] data sets using several recent real-time object detectors with distinct architectures. Specifically, we assess EnDet on YOLOv8 [54] and YOLOv10 [55], which represent the latest advancements in one-stage anchor-free detection, as well as RT-DETR [56], a transformer-based detection

framework. These detectors feature architectural innovations, such as decoupled heads, enhanced feature aggregation, and attention-based modules, representing the current SOTA in real-time object detection.

In addition to the baseline detectors, we compare EnDet with recent UOD methods, including CEH-YOLO [57], FBDPN [58], and PE-Transformer [59]. The experimental results are summarized in Table X. As shown, EnDet consistently achieves higher or comparable performance when integrated with these modern detection frameworks, outperforming both the baseline detectors and the latest underwater detection methods. Notably, EnDet improves detection performance regardless of the underlying detector architecture, demonstrating its effectiveness and flexibility across diverse real-time detection scenarios.

V. DISCUSSION

We explore the limitations of our method by assessing whether various synthesized underwater image data sets could negatively

affect detection performance. Table XI presents experiments in which the synthetic underwater images in $\mathcal{D}_s$ used for the UIE task are replaced with images from LNRUD [60] and UWCNN [61]. The results clearly indicate that the SYREA data set achieves superior performance compared to LNRUD and UWCNN. We attribute this advantage to SYREA's broader diversity and its closer representation of real-world underwater conditions. Specifically, SYREA contains over 20 000 image pairs synthesized from a wide variety of indoor and outdoor terrestrial images, capturing a richer range of underwater degradation and scene variability. This diversity and realism significantly enhance the network's generalization to various underwater environments, thus improving both UIE and UOD performance. In contrast, UWCNN's limited scope (only 1449 indoor images) and LNRUD's less realistic synthetic images restrict the diversity and representativeness of training data, limiting their effectiveness for robust model training and generalization. This demonstrates that the selection of synthetic underwater images plays a critical role not only in UIE capability but also in overall UOD performance.

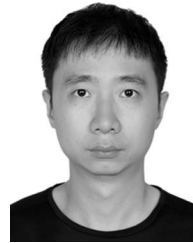
VI. CONCLUSION

In this study, we addressed the challenges posed by the complex underwater environment on deep learning-based object detection by proposing EnDet, a novel framework that jointly trains UOD and UIE in a multitask architecture, featuring with progressively training strategy and novel domain-adaptation mechanisms. Extensive experiments demonstrate that EnDet achieves SOTA performance on UOD tasks across multiple underwater data sets while maintaining a balance between detection and enhancement tasks. In addition, it shows strong adaptability to new environments and exceptional potential for real-time underwater detection. This work marks a significant advancement in improving detection accuracy under challenging underwater conditions. Furthermore, it establishes a foundation for future research aimed at integrating image enhancement with high-level vision tasks.

REFERENCES

- [1] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.
- [2] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 779–788.
- [3] G. Jocher, A. Chaurasia, and J. Qiu, "YOLOv5," Zenodo, 2023, doi: [10.5281/zenodo.3908559](https://doi.org/10.5281/zenodo.3908559).
- [4] W.-H. Lin, J.-X. Zhong, S. Liu, T. Li, and G. Li, "Roimix: Proposal-fusion among multiple images for underwater object detection," in *Proc. IEEE Int. Conf. Acoust. Speech Signal Process.*, 2020, pp. 2588–2592.
- [5] P. Song, P. Li, L. Dai, T. Wang, and Z. Chen, "Boosting R-CNN: Reweighting R-CNN samples by RPN's error for underwater object detection," *Neurocomputing*, vol. 530, pp. 150–164, 2023.
- [6] L. Jiang et al., "Underwater species detection using channel sharpening attention," in *Proc. 29th ACM Int. Conf. Multimedia*, 2021, pp. 4259–4267.
- [7] C.-H. Yeh et al., "Lightweight deep neural network for joint learning of underwater object detection and color conversion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6129–6143, Nov. 2022.
- [8] Z. Liu, B. Wang, Y. Li, J. He, and Y. Li, "Unitmodule: A lightweight joint image enhancement module for underwater object detection," *Pattern Recognit.*, vol. 151, 2024, Art. no. 110435.
- [9] Y. Wang et al., "Is underwater image enhancement all object detectors need?," *IEEE J. Ocean. Eng.*, vol. 33, no. 11, pp. 6129–6143, Nov. 2022.
- [10] C. Fu et al., "Rethinking general underwater object detection: Datasets, challenges, and solutions," *Neurocomputing*, vol. 517, pp. 243–256, 2023.
- [11] J. Wen et al., "Enyolo: A real-time framework for domain-adaptive underwater object detection with image enhancement," in *Proc. 2024 IEEE Int. Conf. Robot. Automat.*, 2024, pp. 12613–12619.
- [12] H. Liu, F. Jin, H. Zeng, H. Pu, and B. Fan, "Image enhancement guided object detection in visually degraded scenes," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 10, pp. 14164–14177, Oct. 2024.
- [13] B. Wang, Z. Wang, W. Guo, and Y. Wang, "A dual-branch joint learning network for underwater object detection," *Knowl.-Based Syst.*, vol. 293, p. 111672, 2024.
- [14] N. Cheng, H. Xie, X. Zhu, and H. Wang, "Joint image enhancement learning for marine object detection in natural scene," *Eng. Appl. Artif. Intell.*, vol. 120, 2023, Art. no. 105905.
- [15] Z. Wang, G. Zhang, K. Luan, C. Yi, and M. Li, "Image-fused-guided underwater object detection model based on improved YOLOv7," *Electronics*, vol. 12, no. 19, 2023, Art. no. 4064.
- [16] Y.-W. Chen and S.-C. Pei, "Domain adaptation for underwater image enhancement via content and style separation," *IEEE Access*, vol. 10, pp. 90523–90534, 2022.
- [17] J. Wen et al., "Syreanet: A physically guided underwater image enhancement framework integrating synthetic and real images," in *Proc. 2023 IEEE Int. Conf. Robot. Automat.*, 2023, pp. 5177–5183.
- [18] Z. Wang, L. Shen, M. Xu, M. Yu, K. Wang, and Y. Lin, "Domain adaptation for underwater image enhancement," *IEEE Trans. Image Process.*, vol. 32, pp. 1442–1457, 2023.
- [19] Y. Chen, W. Li, C. Sakaridis, D. Dai, and L. Van Gool, "Domain adaptive faster R-CNN for object detection in the wild," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 3339–3348.
- [20] W. Liu, G. Ren, R. Yu, S. Guo, J. Zhu, and L. Zhang, "Image-adaptive YOLO for object detection in adverse weather conditions," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 1792–1800.
- [21] C. Liu et al., "A dataset and benchmark of underwater object detection for robot picking," in *Proc. 2021 IEEE Int. Conf. Multimedia Expo Workshops*, 2021, pp. 1–6.
- [22] M. Yang and A. Sowmya, "An underwater color image quality evaluation metric," *IEEE Trans. Image Process.*, vol. 24, no. 12, pp. 6062–6071, Dec. 2015.
- [23] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2014, pp. 580–587.
- [24] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2015, pp. 1440–1448.
- [25] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. 14th Eur. Conf. Comput. Vis.*, Amsterdam, The Netherlands, 2016, pp. 21–37.
- [26] G. Jocher et al., "Ultralytics/ultralytics: YOLO," Zenodo, 2023, doi: [10.5281/zenodo.7347926](https://doi.org/10.5281/zenodo.7347926).
- [27] M. A. Syariz, C.-H. Lin, M. V. Nguyen, L. M. Jaelani, and A. C. Blanco, "Waternet: A convolutional neural network for chlorophyll-a concentration retrieval," *Remote Sens.*, vol. 12, no. 12, 2020, Art. no. 1966.
- [28] L. Dai, H. Liu, P. Song, and M. Liu, "A gated cross-domain collaborative network for underwater object detection," *Pattern Recognit.*, vol. 149, 2024, Art. no. 110222.
- [29] Y. Wang, J. Guo, H. Gao, and H. Yue, "Uiec²-net: CNN-based underwater image enhancement using two color space," *Signal Processing: Image Commun.*, vol. 96, 2021, Art. no. 116250.
- [30] C. Li, S. Anwar, and F. Porikli, "Underwater scene prior inspired deep underwater image and video enhancement," *Pattern Recognit.*, vol. 98, 2020, Art. no. 107038.
- [31] J. Zhou, L. Pang, D. Zhang, and W. Zhang, "Underwater image enhancement method via multi-interval subhistogram perspective equalization," *IEEE J. Ocean. Eng.*, vol. 48, no. 2, pp. 474–488, Apr. 2023.
- [32] X. Fu and X. Cao, "Underwater image enhancement with global-local networks and compressed-histogram equalization," *Signal Process.: Image Commun.*, vol. 86, 2020, Art. no. 115892.
- [33] H.-H. Chang, C.-Y. Cheng, and C.-C. Sung, "Single underwater image restoration based on depth estimation and transmission compensation," *IEEE J. Ocean. Eng.*, vol. 44, no. 4, pp. 1130–1149, Oct. 2019.
- [34] G. Zhang, C. Li, J. Yan, and Y. Zheng, "ULD-cycleGAN: An underwater light field and depth map-optimized cycleGAN for underwater image enhancement," *IEEE J. Ocean. Eng.*, vol. 49, no. 4, pp. 1275–1288, Oct. 2024.

- [35] S. Huang, K. Wang, H. Liu, J. Chen, and Y. Li, "Contrastive semi-supervised learning for underwater image restoration via reliable bank," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 18145–18155.
- [36] J. Wen et al., "Waterformer: A global-local transformer for underwater image enhancement with environment adaptor," *IEEE Robot. Automat. Mag.*, vol. 31, no. 1, pp. 29–40, 2024.
- [37] F. Xiao, F. Yuan, Y. Huang, and E. Cheng, "Turbid underwater image enhancement based on parameter-tuned stochastic resonance," *IEEE J. Ocean. Eng.*, vol. 48, no. 1, pp. 127–146, Mar. 2024.
- [38] Y.-J. Li et al., "Cross-domain adaptive teacher for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 7581–7590.
- [39] H. Liu, P. Song, and R. Ding, "Towards domain generalization in underwater object detection," in *Proc. 2020 IEEE Int. Conf. Image Process.*, 2020, pp. 1971–1975.
- [40] J. Yoo, Y. Uh, S. Chun, B. Kang, and J.-W. Ha, "Photorealistic style transfer via wavelet transforms," in *Proc. Int. Conf. Comput. Vis.*, 2019, pp. 9035–9044.
- [41] Y.-C. Liu et al., "Unbiased teacher for semi-supervised object detection," in *Proc. Int. Conf. Learn. Representations*, 2021.
- [42] B. Sun and K. Saenko, "Deep coral: Correlation alignment for deep domain adaptation," in *Proc. Workshops Comput. Vis.*, Amsterdam, The Netherlands, 2016, pp. 443–450.
- [43] P. Wang, K. Han, X.-S. Wei, L. Zhang, and L. Wang, "Contrastive learning based hybrid networks for long-tailed image classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 943–952.
- [44] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 2961–2969.
- [45] G. Buchsbaum, "A spatial processor model for object colour perception," *J. Franklin Inst.*, vol. 310, no. 1, pp. 1–26, 1980.
- [46] M. Pedersen, J. Bruslund Haurum, R. Gade, and T. B. Moeslund, "Detection of marine animals in a new underwater dataset with varying visibility," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, 2019, pp. 18–26.
- [47] T. Falk et al., "U-net: Deep learning for cell counting, detection, and morphometry," *Nature Methods*, vol. 16, no. 1, pp. 67–70, 2019.
- [48] Y. Yao, L. Rosasco, and A. Caponnetto, "On early stopping in gradient descent learning," *Constructive Approximation*, vol. 26, no. 2, pp. 289–315, 2007.
- [49] M. J. Islam, Y. Xia, and J. Sattar, "Fast underwater image enhancement for improved visual perception," *IEEE Robot. Automat. Lett.*, vol. 5, no. 2, pp. 3227–3234, Apr. 2020.
- [50] Q. Qin, K. Chang, M. Huang, and G. Li, "Denet: Detection-driven enhancement network for object detection under adverse weather conditions," in *Proc. Asian Conf. Comput. Vis.*, 2022, pp. 2813–2829.
- [51] K. Panetta, C. Gao, and S. Agaian, "Human-visual-system-inspired underwater image quality measures," *IEEE J. Ocean. Eng.*, vol. 41, no. 3, pp. 541–551, Jul. 2016.
- [52] Z. Fu, X. Fu, Y. Huang, and X. Ding, "Twice mixing: A rank learning based quality assessment approach for underwater image enhancement," *Signal Process.: Image Commun.*, 2022, Art. no. 116622.
- [53] C. Guo et al., "Underwater ranker: Learn which is better and how to be better," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 702–709.
- [54] G. Jocher et al., "Ultralytics YOLOv8," Zenodo, 2023, doi: [10.5281/zenodo.7347926](https://doi.org/10.5281/zenodo.7347926).
- [55] A. Wang et al., "Yolov10: Real-time end-to-end object detection," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 107984–108011.
- [56] Y. Zhao et al., "Detrs beat yolos on real-time object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024.
- [57] J. Feng and T. Jin, "Ceh-yolo: A composite enhanced YOLO-based model for underwater object detection," *Ecological Informat.*, vol. 82, 2024, Art. no. 102758.
- [58] X. Ji, S. Chen, L.-Y. Hao, J. Zhou, and L. Chen, "FBDPN: CNN-transformer hybrid feature boosting and differential pyramid network for underwater object detection," *Expert Syst. Appl.*, vol. 256, 2024, Art. no. 124978.
- [59] J. Gao, Y. Zhang, X. Geng, H. Tang, and U. A. Bhatti, "Pe-transformer: Path enhanced transformer for improving underwater object detection," *Expert Syst. Appl.*, vol. 246, 2024, Art. no. 123253.
- [60] T. Ye, S. Chen, Y. Liu, Y. Ye, E. Chen, and Y. Li, "Underwater light field retention: Neural rendering for underwater imaging," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 488–497.
- [61] C. Li et al., "An underwater image enhancement benchmark dataset and beyond," *IEEE Trans. Image Process.*, vol. 29, pp. 4376–4389, 2019.



Junjie Wen (Member, IEEE) received the B.Sc. degree in automotive engineering from the Dalian University of Technology, Dalian, China, in 2013, the M.Sc. degree in mechanical engineering from Tsinghua University, Beijing, China, in 2016. He is currently working toward the Ph.D. degree in mechanical and automation engineering Chinese University of Hong Kong, Hong Kong.

His research interests include image restoration, restoration oriented detection, vision-based navigation, etc.



Guidong Yang received the B.Eng. degree in mechanical and automation engineering and the M.Eng. degree in vehicle engineering from Shanghai Jiao Tong University, Shanghai, China, in 2018 and 2021, respectively, and the M.Sc. degree in mechanical engineering from Politecnico di Milano, Milan, Italy, in 2021. He is currently working toward the Ph.D. degree in mechanical and automation engineering with the Chinese University of Hong Kong, Hong Kong.

His current research interests include image enhancement, multiview stereo and unmanned systems.



Benyun Zhao received the M.Sc. degree in mechanical and automation engineering in 2021 from the Chinese University of Hong Kong (CUHK), Hong Kong, where he is currently working toward the Ph.D. degree in mechanical and automation engineering.

He was a Research Assistant with CUHK and the Hong Kong Center of Logistics Robotics (HKCLR) from 2021 to 2022. He is currently a Visiting Student with HiPeRLab, UC Berkeley, Berkeley, CA, USA. His current research interests include object detection, semantic segmentation, and 3-D scene understanding.



Lei Lei received the B.E. degree in naval architecture and ocean engineering from the Department of Ship Engineering, Harbin Engineering University, Harbin, China, in 2016, the M.E. degree in mechanical engineering from the Department of mechanical science and engineering, Huazhong University of Science and Engineering, Wuhan, China in 2019, and the Ph.D. degree in systems engineering from the Department of Systems Engineering, City University of Hong Kong, Hong Kong, in 2024.

He is currently a Postdoctoral Fellow with the Department of Mechanical and Automation Engineering, The Chinese University of Hong Kong, Hong Kong. His research interests include underwater unmanned systems and ocean Big Data.



Zhi Gao received the B.E. and Ph.D. degrees from Wuhan University, Wuhan, China, in 2002 and 2007, respectively.

In 2008, he joined the Interactive and Digital Media Institute, National University of Singapore (NUS), Singapore, as a Research Fellow and the Project Manager. In 2014, he joined the Temasek Laboratories, NUS (TL@NUS), as a Research Scientist and the Principal Investigator. He is currently a Full Professor with the School of Remote Sensing and Information Engineering, Wuhan University. He has authored or

coauthored more than 90 academic articles, which have been published in *IJCV*, *IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE*, *IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING*, *IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS*, and other top journals.

Dr. Gao was the recipient of the prestigious "National Plan for Young Talents" Award and the Hubei Province Funds for Distinguished Young Scientists. In addition, he is a "Chutian Scholar" Distinguished Professor in Hubei. He is an Associate Editor for the journal *Unmanned Systems*.



international journals and coauthored a book in *Green Building and Renewable Application Areas*.

Dr. Chen was the recipient of the DECRA Fellow in the Australian Research Council and Fulbright Scholar in the Lawrence Berkeley National Laboratory. In addition, he serves as the Editorial Board Member of *Buildings*, *Energies*, and *Advances in Applied Energy*.

Xi Chen (Member, IEEE) is currently a Research Assistant Professor of mechanical and automation engineering with the Chinese University of Hong Kong, Hong Kong. He has over 10-year experience in sustainable building technology related to the urban energy systems, renewable application in buildings and built environment modelling, and has led or managed multiple research projects including ARC, MOST, RGC, and consultancy projects with the local government and industry. He has authored or coauthored more than 40 papers in peer-reviewed



research monographs in control theory and applications, unmanned systems and financial market modeling. His current research focuses on unmanned systems and their applications.

Dr. Chen is a Fellow of the Academy of Engineering, Singapore. He had served on the editorial boards of a dozen international journals including *AUTOMATICA* and *IEEE TRANSACTIONS ON AUTOMATIC CONTROL*. He is currently serving as the Editor-in-Chief of *Unmanned Systems*, and an Editor of the *International Journal of Robust and Nonlinear Control*.

Ben M. Chen (Fellow, IEEE) is currently a Professor of mechanical and automation engineering with the Chinese University of Hong Kong (CUHK), Hong Kong. He was a Provost's Chair Professor with the Department of Electrical and Computer Engineering, National University of Singapore, Singapore, before joining CUHK, in 2018. From 1992 to 1993, he was an Assistant Professor with the Department of Electrical Engineering, State University of New York, Stony Brook, NY, USA. He has authored/co-authored hundreds of journal and conference articles, and a dozen